

20. Let p be a prime and let n be a positive integer. Show that if x is an element of the group G such that $x^{p^n} = 1$ then $|x| = p^m$ for some $m \leq n$.

Put $k = |x|$. Since k divides p^n , $k = p^m$, $m \leq n$

+ More precisely, by the Bezout's equation, $\gcd(k, p^n) = ks + p^nt$ for some $s, t \in \mathbb{Z}$.

Then, $x^{\gcd(k, p^n)} = x^{ks + p^nt} = x^{ks} \cdot x^{p^nt} = 1 \cdot 1 = 1$,

Thus $\gcd(k, p^n) \mid k$ and $k \leq \gcd(k, p^n)$, $k = \gcd(k, p^n)$.

Therefore $k \mid p^n$, p is prime, $k = p^m$ for some $m \leq n$.

21. Let p be an odd prime and let n be a positive integer. Use the Binomial Theorem to show that $(1+p)^{p^{n-1}} \equiv 1 \pmod{p^n}$ but $(1+p)^{p^{n-2}} \not\equiv 1 \pmod{p^n}$. Deduce that $1+p$ is an element of order p^{n-1} in the multiplicative group $(\mathbb{Z}/p^n\mathbb{Z})^\times$.

$$(1+p)^{p^{n-1}} = \sum_{k=0}^{p^{n-1}} \binom{p^{n-1}}{k} \cdot p^k$$

$$= \mathbb{Z}_{p^n}^\times$$

observe that: for $k \geq 1$, $\binom{p^{n-1}}{k} \cdot p^k = \frac{p^{n-1}!}{k!(p^{n-1}-k)!} \cdot p^k = \frac{(p^{n-1}-1)!}{k!(p^{n-1}-k)!} \cdot p^{n+k-1} \equiv 0 \pmod{p^n}$.
Therefore, clearly $k=0$, $1 \equiv 1 \pmod{p^n}$.

$$(1+p)^{p^{n-1}} \pmod{p^n} = \sum_{k=0}^{p^{n-1}} \binom{p^{n-1}}{k} \cdot p^k = 1 + \sum_{k=1}^{p^{n-1}} 0 = 1. \text{ But,}$$

if $(1+p)^{p^{n-2}} \equiv 1 \pmod{p^n}$, then $(1+p)^{p^{n-1}} \equiv 1+p \pmod{p^n} = 1+p$, Contradiction.

Now, $(1+p)^{p^{n-1}} = 1$ in $\mathbb{Z}_{p^n}^\times$. And, if $(1+p)^m = 1$, $m \mid p^{n-1}$

Thus $m = p^{n-1}, p^{n-2}, \dots, p^1$, But similarly in the case of $(1+p)^{p^{n-2}}$,

- 22.** Let n be an integer ≥ 3 . Use the Binomial Theorem to show that $(1 + 2^2)^{2^{n-2}} \equiv 1 \pmod{2^n}$ but $(1 + 2^2)^{2^{n-3}} \not\equiv 1 \pmod{2^n}$. Deduce that 5 is an element of order 2^{n-2} in the multiplicative group $(\mathbb{Z}/2^n\mathbb{Z})^\times$.

- 23.** Show that $(\mathbb{Z}/2^n\mathbb{Z})^\times$ is not cyclic for any $n \geq 3$. [Find two distinct subgroups of order 2.]

$$\mathbb{Z}_{2^n}^\times$$

24. Let G be a finite group and let $x \in G$.

(a) Prove that if $g \in N_G(\langle x \rangle)$ then $gxg^{-1} = x^a$ for some $a \in \mathbb{Z}$.

(b) Prove conversely that if $gxg^{-1} = x^a$ for some $a \in \mathbb{Z}$ then $g \in N_G(\langle x \rangle)$. [Show first that $gx^k g^{-1} = (gxg^{-1})^k = x^{ak}$ for any integer k , so that $g\langle x \rangle g^{-1} \leq \langle x \rangle$. If x has order n , show the elements $gx^i g^{-1}$, $i = 0, 1, \dots, n-1$ are distinct, so that $|g\langle x \rangle g^{-1}| = |\langle x \rangle| = n$ and conclude that $g\langle x \rangle g^{-1} = \langle x \rangle$.]

Note that this cuts down some of the work in computing normalizers of cyclic subgroups since one does not have to check $ghg^{-1} \in \langle x \rangle$ for every $h \in \langle x \rangle$.

a). Since $g \in N_G(\langle x \rangle) \Leftrightarrow g\langle x \rangle g^{-1} = \langle x \rangle$,

$gxg^{-1} \in g\langle x \rangle g^{-1} = \langle x \rangle$. Thus, for some $a \in \mathbb{Z}$, $gxg^{-1} = x^a$.

b) Conversely, suppose $g \in G$ satisfies $gxg^{-1} = x^a$, for some $a \in \mathbb{Z}$.

Let $k \in \mathbb{Z}$. Then, $(gxg^{-1})^k = gxg^{-1} \cdot gxg^{-1} \cdots gxg^{-1}$
 $= gx^k g^{-1}$
 $= (x^a)^k = x^{ak}$.

Therefore, $g\langle x \rangle g^{-1} \subseteq \langle x \rangle$, and it is subgroup, because $\langle x \rangle$ is a subgroup.

Moreover, let $n = |x|$. Then, $gx^0 g^{-1}, gx^1 g^{-1}, \dots, gx^{n-1} g^{-1}$ are distinct,

thus $|g\langle x \rangle g^{-1}| = |\langle x \rangle| = n$. Consequently $g\langle x \rangle g^{-1} = \langle x \rangle$, or $g \in N_G(\langle x \rangle)$

Summary. If G is finite, then

$g \in N_G(\langle x \rangle) \Leftrightarrow gxg^{-1} = x^a$ for some $a \in \mathbb{Z}$

$\Leftrightarrow gxg^{-1} \in \langle x \rangle$

25. Let G be a cyclic group of order n and let k be an integer relatively prime to n . Prove that the map $x \mapsto x^k$ is surjective. Use Lagrange's Theorem (Exercise 19, Section 1.7) to prove the same is true for any finite group of order n . (For such k each element has a k^{th} root in G . It follows from Cauchy's Theorem in Section 3.2 that if k is not relatively prime to the order of G then the map $x \mapsto x^k$ is not surjective.)

$$x \mapsto x^k \text{ is onto} \Leftrightarrow \text{for any } 0 \leq m \leq n-1, \text{ there exists } 0 \leq r \leq n-1 \text{ s.t. } (x^n)^k = x^{rk} = x^m$$

$$\Leftrightarrow rk \equiv m \pmod{n}$$

$$\Leftrightarrow \gcd(n, k) \mid m$$

Since $\gcd(n, k) = 1$, it is true.

Now, let G be a finite group of order n , not necessarily cyclic.

Let $y \in G$ be given. Then, $|y|$ divides n . Denote $m = |y|$. Since $m \mid n$, $\gcd(m, k) = 1$.

Then, there is a solution of $tk \equiv 1 \pmod{m}$.

$$\text{thus } y = (y^t)^k = y^{tk}.$$

Note: $ax \equiv b \pmod{n}$ has solution
 $\Leftrightarrow \gcd(a, n) \mid b$, number of d.

$$\left(\begin{array}{l} tk \equiv 1 \pmod{m} \Leftrightarrow tk - 1 = mZ \quad (Z \in \mathbb{Z}) \\ \Rightarrow y^{tk} = y^{mZ+1} = y. \end{array} \right.$$

26. Let Z_n be a cyclic group of order n and for each integer a let

$$\sigma_a : Z_n \rightarrow Z_n \quad \text{by} \quad \sigma_a(x) = x^a \text{ for all } x \in Z_n.$$

(a) Prove that σ_a is an automorphism of Z_n if and only if a and n are relatively prime (automorphisms were introduced in Exercise 20, Section 1.6).

(b) Prove that $\sigma_a = \sigma_b$ if and only if $a \equiv b \pmod{n}$.

(c) Prove that every automorphism of Z_n is equal to σ_a for some integer a .

(d) Prove that $\sigma_a \circ \sigma_b = \sigma_{ab}$. Deduce that the map $\bar{a} \mapsto \sigma_a$ is an isomorphism of $(\mathbb{Z}/n\mathbb{Z})^\times$ onto the automorphism group of Z_n (so $\text{Aut}(Z_n)$ is an abelian group of order $\phi(n)$).

a). Whether a is relatively prime or not, $\sigma_a(xy) = (xy)^a = x^a y^a = \sigma_a(x) \cdot \sigma_a(y)$, homomorphism.

And, since Z_n is finite, σ_a is bijection $\Leftrightarrow \sigma_a$ onto or 1-1. Z_n abelian.

Since $Z_n = \langle 1 \rangle$, if $\gcd(a, n) = 1$, σ_a is onto.

If $\gcd(a, n) \neq 1$, put this to $d = \gcd(a, n)$. Now, $\frac{a}{d}, \frac{n}{d} \in \mathbb{N}$,

$$\sigma_a\left(\frac{d}{a}\right) =$$

1. Prove that if H is a subgroup of G then $\langle H \rangle = H$.
2. Prove that if A is a subset of B then $\langle A \rangle \leq \langle B \rangle$. Give an example where $A \subseteq B$ with $A \neq B$ but $\langle A \rangle = \langle B \rangle$.
3. Prove that if H is an abelian subgroup of a group G then $\langle H, Z(G) \rangle$ is abelian. Give an explicit example of an abelian subgroup H of a group G such that $\langle H, C_G(H) \rangle$ is not abelian.
4. Prove that if H is a subgroup of G then H is generated by the set $H - \{1\}$.

1. $H \subseteq \langle H \rangle$ is trivial. Let $x \in \langle H \rangle$ be given. Then, $x = g_1 \cdots g_n$ for some $g_i \in H$, and therefore $x \in H$.

5. Prove that the subgroup generated by any two distinct elements of order 2 in S_3 is all of S_3 .
6. Prove that the subgroup of S_4 generated by $(1\ 2)$ and $(1\ 2)(3\ 4)$ is a noncyclic group of order 4.
7. Prove that the subgroup of S_4 generated by $(1\ 2)$ and $(1\ 3)(2\ 4)$ is isomorphic to the dihedral group of order 8.
8. Prove that $S_4 = \langle (1\ 2\ 3\ 4), (1\ 2\ 4\ 3) \rangle$.

9. Prove that $SL_2(\mathbb{F}_3)$ is the subgroup of $GL_2(\mathbb{F}_3)$ generated by $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$. [Recall from Exercise 9 of Section 1 that $SL_2(\mathbb{F}_3)$ is the subgroup of matrices of determinant 1. You may assume this subgroup has order 24 — this will be an exercise in Section 3.2.]
10. Prove that the subgroup of $SL_2(\mathbb{F}_3)$ generated by $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ is isomorphic to the quaternion group of order 8. [Use a presentation for Q_8 .]
11. Show that $SL_2(\mathbb{F}_3)$ and S_4 are two nonisomorphic groups of order 24.
12. Prove that the subgroup of upper triangular matrices in $GL_3(\mathbb{F}_2)$ is isomorphic to the dihedral group of order 8 (cf. Exercise 16, Section 1). [First find the order of this subgroup.]

- 14.** A group H is called *finitely generated* if there is a finite set A such that $H = \langle A \rangle$.
- (a) Prove that every finite group is finitely generated.
 - (b) Prove that $\mathbb{Z}$ is finitely generated.
 - (c) Prove that every finitely generated subgroup of the additive group $\mathbb{Q}$ is cyclic. [If H is a finitely generated subgroup of $\mathbb{Q}$, show that $H \leq \langle \frac{1}{k} \rangle$, where k is the product of all the denominators which appear in a set of generators for H .]
 - (d) Prove that $\mathbb{Q}$ is not finitely generated.

- 16.** A subgroup M of a group G is called a *maximal subgroup* if $M \neq G$ and the only subgroups of G which contain M are M and G .
- (a) Prove that if H is a proper subgroup of the finite group G then there is a maximal subgroup of G containing H .
 - (b) Show that the subgroup of all rotations in a dihedral group is a maximal subgroup.
 - (c) Show that if $G = \langle x \rangle$ is a cyclic group of order $n \geq 1$ then a subgroup H is maximal if and only if $H = \langle x^p \rangle$ for some prime p dividing n .

- 17.** This is an exercise involving Zorn's Lemma (see Appendix I) to prove that every nontrivial finitely generated group possesses maximal subgroups. Let G be a finitely generated group, say $G = \langle g_1, g_2, \dots, g_n \rangle$, and let $\mathcal{S}$ be the set of all proper subgroups of G . Then $\mathcal{S}$ is partially ordered by inclusion. Let $\mathcal{C}$ be a chain in $\mathcal{S}$.
- (a) Prove that the union, H , of all the subgroups in $\mathcal{C}$ is a subgroup of G .
 - (b) Prove that H is a *proper* subgroup. [If not, each g_i must lie in H and so must lie in some element of the chain $\mathcal{C}$. Use the definition of a chain to arrive at a contradiction.]
 - (c) Use Zorn's Lemma to show that $\mathcal{S}$ has a maximal element (which is, by definition, a maximal subgroup).

- 18.** Let p be a prime and let $Z = \{z \in \mathbb{C} \mid z^{p^n} = 1 \text{ for some } n \in \mathbb{Z}^+\}$ (so Z is the multiplicative group of all p -power roots of unity in $\mathbb{C}$). For each $k \in \mathbb{Z}^+$ let $H_k = \{z \in Z \mid z^{p^k} = 1\}$ (the group of p^k th roots of unity). Prove the following:
- (a) $H_k \leq H_m$ if and only if $k \leq m$
 - (b) H_k is cyclic for all k (assume that for any $n \in \mathbb{Z}^+$, $\{e^{2\pi it/n} \mid t = 0, 1, \dots, n-1\}$ is the set of all n^{th} roots of 1 in $\mathbb{C}$)
 - (c) every proper subgroup of Z equals H_k for some $k \in \mathbb{Z}^+$ (in particular, every proper subgroup of Z is finite and cyclic)
 - (d) Z is not finitely generated.

- 19.** A nontrivial abelian group A (written multiplicatively) is called *divisible* if for each element $a \in A$ and each nonzero integer k there is an element $x \in A$ such that $x^k = a$, i.e., each element has a k^{th} root in A (in additive notation, each element is the k^{th} multiple of some element of A).
- (a) Prove that the additive group of rational numbers, $\mathbb{Q}$, is divisible.
 - (b) Prove that no finite abelian group is divisible.
- 20.** Prove that if A and B are nontrivial abelian groups, then $A \times B$ is divisible if and only if both A and B are divisible groups.

2.5 THE LATTICE OF SUBGROUPS OF A GROUP

11. Consider the group of order 16 with the following presentation:

$$QD_{16} = \langle \sigma, \tau \mid \sigma^8 = \tau^2 = 1, \sigma\tau = \tau\sigma^3 \rangle$$

(called the *quasidihedral* or *semidihedral* group of order 16). This group has three subgroups of order 8: $\langle \tau, \sigma^2 \rangle \cong D_8$, $\langle \sigma \rangle \cong Z_8$ and $\langle \sigma^2, \sigma\tau \rangle \cong Q_8$ and every proper subgroup is contained in one of these three subgroups. Fill in the missing subgroups in the lattice of all subgroups of the quasidihedral group on the following page, exhibiting each subgroup with at most two generators. (This is another example of a nonplanar lattice.)

EXERCISES

Let G and H be groups.

1. Let $\varphi : G \rightarrow H$ be a homomorphism and let E be a subgroup of H . Prove that $\varphi^{-1}(E) \leq G$ (i.e., the preimage or pullback of a subgroup under a homomorphism is a subgroup). If $E \trianglelefteq H$ prove that $\varphi^{-1}(E) \trianglelefteq G$. Deduce that $\ker \varphi \trianglelefteq G$.
2. Let $\varphi : G \rightarrow H$ be a homomorphism of groups with kernel K and let $a, b \in \varphi(G)$. Let $X \in G/K$ be the fiber above a and let Y be the fiber above b , i.e., $X = \varphi^{-1}(a)$, $Y = \varphi^{-1}(b)$. Fix an element u of X (so $\varphi(u) = a$). Prove that if $XY = Z$ in the quotient group G/K and w is any member of Z , then there is some $v \in Y$ such that $uv = w$. [Show $u^{-1}w \in Y$.]

3. Let A be an abelian group and let B be a subgroup of A . Prove that A/B is abelian. Give an example of a non-abelian group G containing a proper normal subgroup N such that G/N is abelian.

6. Define $\varphi : \mathbb{R}^\times \rightarrow \{\pm 1\}$ by letting $\varphi(x)$ be x divided by the absolute value of x . Describe the fibers of φ and prove that φ is a homomorphism.
7. Define $\pi : \mathbb{R}^2 \rightarrow \mathbb{R}$ by $\pi((x, y)) = x + y$. Prove that π is a surjective homomorphism and describe the kernel and fibers of π geometrically.
8. Let $\varphi : \mathbb{R}^\times \rightarrow \mathbb{R}^\times$ be the map sending x to the absolute value of x . Prove that φ is a homomorphism and find the image of φ . Describe the kernel and the fibers of φ .
9. Define $\varphi : \mathbb{C}^\times \rightarrow \mathbb{R}^\times$ by $\varphi(a + bi) = a^2 + b^2$. Prove that φ is a homomorphism and find the image of φ . Describe the kernel and the fibers of φ geometrically (as subsets of the plane).

11. Let F be a field and let $G = \left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \mid a, b, c \in F, ac \neq 0 \right\} \leq GL_2(F)$.

(a) Prove that the map $\varphi : \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \mapsto a$ is a surjective homomorphism from G onto $F^\times$ (recall that $F^\times$ is the multiplicative group of nonzero elements in F). Describe the fibers and kernel of φ .

(b) Prove that the map $\psi : \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \mapsto (a, c)$ is a surjective homomorphism from G onto $F^\times \times F^\times$. Describe the fibers and kernel of ψ .

(c) Let $H = \left\{ \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} \mid b \in F \right\}$. Prove that H is isomorphic to the additive group F .

- 14.** Consider the additive quotient group $\mathbb{Q}/\mathbb{Z}$.
- (a) Show that every coset of $\mathbb{Z}$ in $\mathbb{Q}$ contains exactly one representative $q \in \mathbb{Q}$ in the range $0 \leq q < 1$.
 - (b) Show that every element of $\mathbb{Q}/\mathbb{Z}$ has finite order but that there are elements of arbitrarily large order.
 - (c) Show that $\mathbb{Q}/\mathbb{Z}$ is the torsion subgroup of $\mathbb{R}/\mathbb{Z}$ (cf. Exercise 6, Section 2.1).
 - (d) Prove that $\mathbb{Q}/\mathbb{Z}$ is isomorphic to the multiplicative group of root of unity in $\mathbb{C}^\times$.
- 15.** Prove that a quotient of a divisible abelian group by any proper subgroup is also divisible. Deduce that $\mathbb{Q}/\mathbb{Z}$ is divisible (cf. Exercise 19, Section 2.4).

- 16.** Let G be a group, let N be a normal subgroup of G and let $\overline{G} = G/N$. Prove that if $G = \langle x, y \rangle$ then $\overline{G} = \langle \overline{x}, \overline{y} \rangle$. Prove more generally that if $G = \langle S \rangle$ for any subset S of G , then $\overline{G} = \langle \overline{S} \rangle$.

17. Let G be the dihedral group of order 16 (whose lattice appears in Section 2.5):

$$G = \langle r, s \mid r^8 = s^2 = 1, rs = sr^{-1} \rangle$$

and let $\overline{G} = G/\langle r^4 \rangle$ be the quotient of G by the subgroup generated by r^4 (this subgroup is the center of G , hence is normal).

- (a) Show that the order of $\overline{G}$ is 8.
- (b) Exhibit each element of $\overline{G}$ in the form $\overline{s}^a \overline{r}^b$, for some integers a and b .
- (c) Find the order of each of the elements of $\overline{G}$ exhibited in (b).
- (d) Write each of the following elements of $\overline{G}$ in the form $\overline{s}^a \overline{r}^b$, for some integers a and b as in (b): $\overline{rs}$, $\overline{sr^{-2}s}$, $\overline{s^{-1}r^{-1}sr}$.
- (e) Prove that $\overline{H} = \langle \overline{s}, \overline{r}^2 \rangle$ is a normal subgroup of $\overline{G}$ and $\overline{H}$ is isomorphic to the Klein 4-group. Describe the isomorphism type of the complete preimage of $\overline{H}$ in G .
- (f) Find the center of $\overline{G}$ and describe the isomorphism type of $\overline{G}/Z(\overline{G})$.

18. Let G be the quasidihedral group of order 16 (whose lattice was computed in Exercise 11 of Section 2.5):

$$G = \langle \sigma, \tau \mid \sigma^8 = \tau^2 = 1, \sigma\tau = \tau\sigma^3 \rangle$$

and let $\overline{G} = G/\langle \sigma^4 \rangle$ be the quotient of G by the subgroup generated by σ^4 (this subgroup is the center of G , hence is normal).

- (a) Show that the order of $\overline{G}$ is 8.
- (b) Exhibit each element of $\overline{G}$ in the form $\overline{\tau}^a \overline{\sigma}^b$, for some integers a and b .
- (c) Find the order of each of the elements of $\overline{G}$ exhibited in (b).
- (d) Write each of the following elements of $\overline{G}$ in the form $\overline{\tau}^a \overline{\sigma}^b$, for some integers a and b as in (b): $\overline{\sigma\tau}$, $\overline{\tau\sigma^{-2}\tau}$, $\overline{\tau^{-1}\sigma^{-1}\tau\sigma}$.
- (e) Prove that $\overline{G} \cong D_8$.

19. Let G be the modular group of order 16 (whose lattice was computed in Exercise 14 of Section 2.5):

$$G = \langle u, v \mid u^2 = v^8 = 1, vu = uv^5 \rangle$$

and let $\overline{G} = G/\langle v^4 \rangle$ be the quotient of G by the subgroup generated by v^4 (this subgroup is contained in the center of G , hence is normal).

- (a) Show that the order of $\overline{G}$ is 8.
- (b) Exhibit each element of $\overline{G}$ in the form $\overline{u}^a \overline{v}^b$, for some integers a and b .
- (c) Find the order of each of the elements of $\overline{G}$ exhibited in (b).
- (d) Write each of the following elements of $\overline{G}$ in the form $\overline{u}^a \overline{v}^b$, for some integers a and b as in (b): $\overline{vu}$, $\overline{uv^{-2}u}$, $\overline{u^{-1}v^{-1}uv}$.
- (e) Prove that $\overline{G}$ is abelian and is isomorphic to $Z_2 \times Z_4$.

21. Let $G = \mathbb{Z}_4 \times \mathbb{Z}_4$ be given in terms of the following generators and relations:

$$G = \langle x, y \mid x^4 = y^4 = 1, xy = yx \rangle.$$

Let $\overline{G} = G/\langle x^2y^2 \rangle$ (note that every subgroup of the abelian group G is normal).

- (a) Show that the order of $\overline{G}$ is 8.
- (b) Exhibit each element of $\overline{G}$ in the form $\overline{x}^a\overline{y}^b$, for some integers a and b .
- (c) Find the order of each of the elements of $\overline{G}$ exhibited in (b).
- (d) Prove that $\overline{G} \cong \mathbb{Z}_4 \times \mathbb{Z}_2$.

25. (a) Prove that a subgroup N of G is normal if and only if $gNg^{-1} \subseteq N$ for *all* $g \in G$.
- (b) Let $G = GL_2(\mathbb{Q})$, let N be the subgroup of upper triangular matrices with integer entries and 1's on the diagonal, and let g be the diagonal matrix with entries 2,1. Show that $gNg^{-1} \subseteq N$ but g does *not* normalize N .

26. Let $a, b \in G$.

- (a) Prove that the conjugate of the product of a and b is the product of the conjugate of a and the conjugate of b . Prove that the order of a and the order of any conjugate of a are the same.
- (b) Prove that the conjugate of a^{-1} is the inverse of the conjugate of a .
- (c) Let $N = \langle S \rangle$ for some subset S of G . Prove that $N \trianglelefteq G$ if $gSg^{-1} \subseteq N$ for all $g \in G$.
- (d) Deduce that if N is the cyclic group $\langle x \rangle$, then N is normal in G if and only if for each $g \in G$, $gxg^{-1} = x^k$ for some $k \in \mathbb{Z}$.
- (e) Let n be a positive integer. Prove that the subgroup N of G generated by all the elements of G of order n is a normal subgroup of G .

27. Let N be a *finite* subgroup of a group G . Show that $gNg^{-1} \subseteq N$ if and only if $gNg^{-1} = N$. Deduce that $N_G(N) = \{g \in G \mid gNg^{-1} \subseteq N\}$.
28. Let N be a *finite* subgroup of a group G and assume $N = \langle S \rangle$ for some subset S of G . Prove that an element $g \in G$ normalizes N if and only if $gSg^{-1} \subseteq N$.
29. Let N be a *finite* subgroup of G and suppose $G = \langle T \rangle$ and $N = \langle S \rangle$ for some subsets S and T of G . Prove that N is normal in G if and only if $tSt^{-1} \subseteq N$ for all $t \in T$.
30. Let $N \leq G$ and let $g \in G$. Prove that $gN = Ng$ if and only if $g \in N_G(N)$.
31. Prove that if $H \leq G$ and N is a normal subgroup of H then $H \leq N_G(N)$. Deduce that $N_G(N)$ is the largest subgroup of G in which N is normal (i.e., is the join of all subgroups H for which $N \trianglelefteq H$).

32. Prove that every subgroup of Q_8 is normal. For each subgroup find the isomorphism type of its corresponding quotient. [You may use the lattice of subgroups for Q_8 in Section 2.5.]

- 34.** Let $D_{2n} = \langle r, s \mid r^n = s^2 = 1, rs = sr^{-1} \rangle$ be the usual presentation of the dihedral group of order $2n$ and let k be a positive integer dividing n .
- (a) Prove that $\langle r^k \rangle$ is a normal subgroup of D_{2n} .
 - (b) Prove that $D_{2n}/\langle r^k \rangle \cong D_{2k}$.

- 36.** Prove that if $G/Z(G)$ is cyclic then G is abelian. [If $G/Z(G)$ is cyclic with generator $xZ(G)$, show that every element of G can be written in the form $x^a z$ for some integer $a \in \mathbb{Z}$ and some element $z \in Z(G)$.]

- 37.** Let A and B be groups. Show that $\{(a, 1) \mid a \in A\}$ is a normal subgroup of $A \times B$ and the quotient of $A \times B$ by this subgroup is isomorphic to B .
- 38.** Let A be an abelian group and let D be the (diagonal) subgroup $\{(a, a) \mid a \in A\}$ of $A \times A$. Prove that D is a normal subgroup of $A \times A$ and $(A \times A)/D \cong A$.
- 39.** Suppose A is the non-abelian group S_3 and D is the diagonal subgroup $\{(a, a) \mid a \in A\}$ of $A \times A$. Prove that D is not normal in $A \times A$.

40. Let G be a group, let N be a normal subgroup of G and let $\overline{G} = G/N$. Prove that $\overline{x}$ and $\overline{y}$ commute in $\overline{G}$ if and only if $x^{-1}y^{-1}xy \in N$. (The element $x^{-1}y^{-1}xy$ is called the *commutator* of x and y and is denoted by $[x, y]$.)
41. Let G be a group. Prove that $N = \langle x^{-1}y^{-1}xy \mid x, y \in G \rangle$ is a normal subgroup of G and G/N is abelian (N is called the *commutator subgroup* of G).
42. Assume both H and K are normal subgroups of G with $H \cap K = 1$. Prove that $xy = yx$ for all $x \in H$ and $y \in K$. [Show $x^{-1}y^{-1}xy \in H \cap K$.]

4. Show that if $|G| = pq$ for some primes p and q (not necessarily distinct) then either G is abelian or $Z(G) = 1$. [See Exercise 36 in Section 1.]

5. Let H be a subgroup of G and fix some element $g \in G$.

(a) Prove that gHg^{-1} is a subgroup of G of the same order as H .

(b) Deduce that if $n \in \mathbb{Z}^+$ and H is the unique subgroup of G of order n then $H \trianglelefteq G$.

6. Let $H \leq G$ and let $g \in G$. Prove that if the right coset Hg equals *some* left coset of H in G then it equals the left coset gH and g must be in $N_G(H)$.
7. Let $H \leq G$ and define a relation $\sim$ on G by $a \sim b$ if and only if $b^{-1}a \in H$. Prove that $\sim$ is an equivalence relation and describe the equivalence class of each $a \in G$. Use this to prove Proposition 4.
8. Prove that if H and K are finite subgroups of G whose orders are relatively prime then $H \cap K = 1$.

- 10.** Suppose H and K are subgroups of finite index in the (possibly infinite) group G with $|G : H| = m$ and $|G : K| = n$. Prove that $\text{l.c.m.}(m, n) \leq |G : H \cap K| \leq mn$. Deduce that if m and n are relatively prime then $|G : H \cap K| = |G : H| \cdot |G : K|$.
- 11.** Let $H \leq K \leq G$. Prove that $|G : H| = |G : K| \cdot |K : H|$ (do not assume G is finite).

- 12.** Let $H \leq G$. Prove that the map $x \mapsto x^{-1}$ sends each left coset of H in G onto a right coset of H and gives a bijection between the set of left cosets and the set of right cosets of H in G (hence the number of left cosets of H in G equals the number of right cosets).

- 14.** Prove that S_4 does not have a normal subgroup of order 8 or a normal subgroup of order 3.
- 15.** Let $G = S_n$ and for fixed $i \in \{1, 2, \dots, n\}$ let G_i be the stabilizer of i . Prove that $G_i \cong S_{n-1}$.

- 16.** Use Lagrange's Theorem in the multiplicative group $(\mathbb{Z}/p\mathbb{Z})^\times$ to prove *Fermat's Little Theorem*: if p is a prime then $a^p \equiv a \pmod{p}$ for all $a \in \mathbb{Z}$.

17. Let p be a prime and let n be a positive integer. Find the order of $\bar{p}$ in $(\mathbb{Z}/(p^n-1)\mathbb{Z})^\times$ and deduce that $n \mid \varphi(p^n - 1)$ (here φ is Euler's function).

- 18.** Let G be a finite group, let H be a subgroup of G and let $N \trianglelefteq G$. Prove that if $|H|$ and $|G : N|$ are relatively prime then $H \leq N$.
- 19.** Prove that if N is a normal subgroup of the finite group G and $(|N|, |G : N|) = 1$ then N is the unique subgroup of G of order $|N|$.
- 20.** If A is an abelian group with $A \trianglelefteq G$ and B is any subgroup of G prove that $A \cap B \trianglelefteq AB$.

- 22.** Use Lagrange's Theorem in the multiplicative group $(\mathbb{Z}/n\mathbb{Z})^\times$ to prove *Euler's Theorem*: $a^{\varphi(n)} \equiv 1 \pmod{n}$ for every integer a relatively prime to n , where φ denotes Euler's φ -function.

3. Prove that if H is a normal subgroup of G of prime index p then for all $K \leq G$ either
- (i) $K \leq H$ or
 - (ii) $G = HK$ and $|K : K \cap H| = p$.
4. Let C be a normal subgroup of the group A and let D be a normal subgroup of the group B . Prove that $(C \times D) \trianglelefteq (A \times B)$ and $(A \times B)/(C \times D) \cong (A/C) \times (B/D)$.

7. Let M and N be normal subgroups of G such that $G = MN$. Prove that $G/(M \cap N) \cong (G/M) \times (G/N)$. [Draw the lattice.]

8. Let p be a prime and let G be the group of p -power roots of 1 in $\mathbb{C}$ (cf. Exercise 18, Section 2.4). Prove that the map $z \mapsto z^p$ is a surjective homomorphism. Deduce that G is isomorphic to a proper quotient of itself.
9. Let p be a prime and let G be a group of order $p^a m$, where p does not divide m . Assume P is a subgroup of G of order p^a and N is a normal subgroup of G of order $p^b n$, where p does not divide n . Prove that $|P \cap N| = p^b$ and $|PN/N| = p^{a-b}$. (The subgroup P of G is called a *Sylow p -subgroup* of G . This exercise shows that the intersection of any Sylow p -subgroup of G with a normal subgroup N is a Sylow p -subgroup of N .)
10. Generalize the preceding exercise as follows. A subgroup H of a finite group G is called a *Hall subgroup* of G if its index in G is relatively prime to its order: $(|G : H|, |H|) = 1$. Prove that if H is a Hall subgroup of G and $N \trianglelefteq G$, then $H \cap N$ is a Hall subgroup of N and HN/N is a Hall subgroup of G/N .